

# Certified safe set on the active $\pm 4$ ODD contract

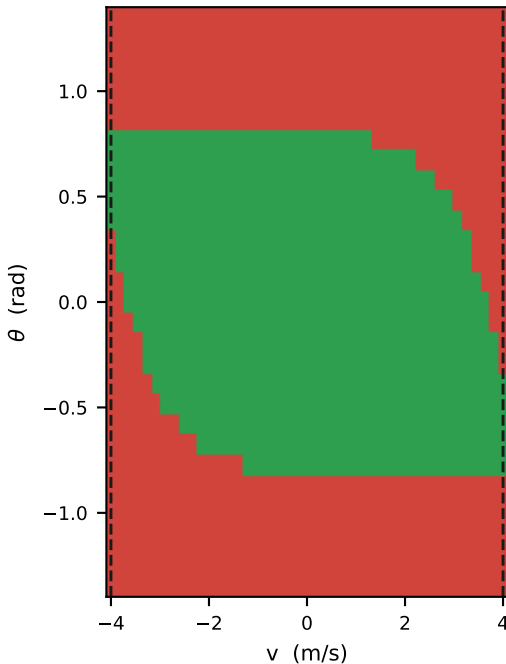
$v \in [-4, 4]$  m/s,  $|\dot{\psi}| \leq 4$  rad/s — read from the locked contract (velocity tier: policy)

The velocity edges are the CONTRACT, not the dynamics. The four-state model has no speed-dependent term — pitch acceleration is identical at  $v = 0$  through 10 m/s — so along  $v$  the value function is exactly the ODD margin ramp and the certified set ends wherever the declared bound is placed. What the solve genuinely determines is the boundary in pitch, pitch rate, and the  $|v\dot{\psi}| \leq a_{tip}$  roll coupling: those edges are dynamics. Widening  $v$  further would only move the wall.

■ certified safe ( $V \geq 0$ )   ■ not certified   - - - declared ODD velocity bound

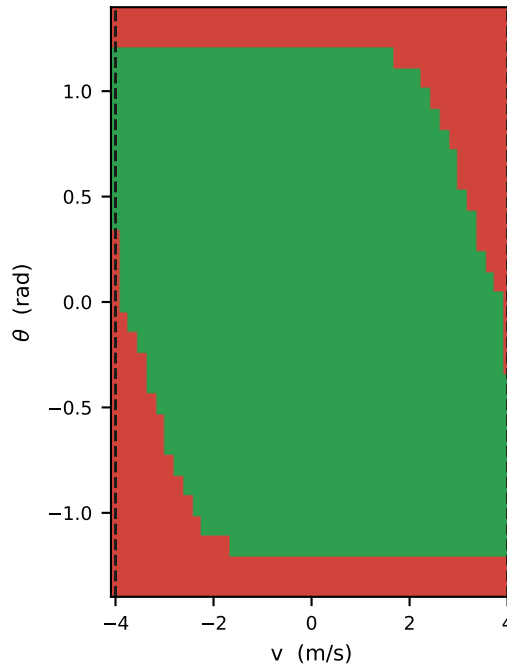
$\mu = 0.3$

25.6% of grid safe · at rest,  $v \in [-3.6, +3.6]$  m/s



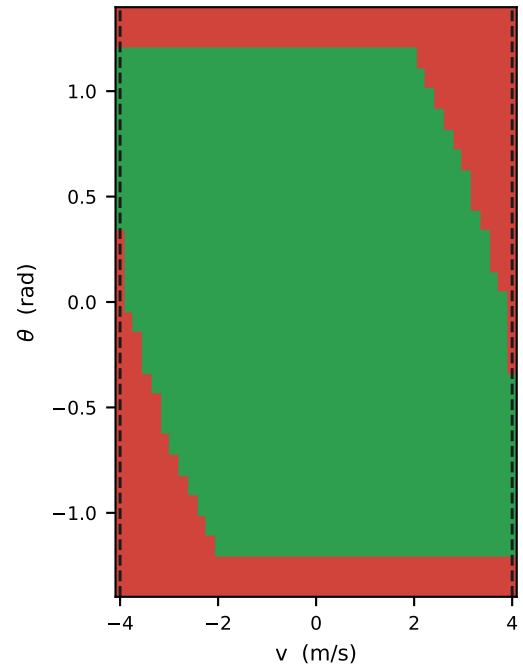
$\mu = 0.6$

43.2% of grid safe · at rest,  $v \in [-3.8, +3.8]$  m/s



$\mu = 1.0$

48.0% of grid safe · at rest,  $v \in [-3.8, +3.8]$  m/s



Slices at  $\dot{\theta} = \dot{\psi} = 0$ . Grid and contract both read from the pinned release, so figure and contract cannot drift.

Not comparable with `make_model_update_figure.py`, which is pinned to the older narrower contract. Reproduce: `python vault/tools/make_odd_safeset_figure.py`